SIMATS ENGINEERING

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ASSESSMENT TOOL III- ASSIGNMENT

COURSE CODE: ITA14 **COURSE NAME: Ethical Hacking**

COURSE OUTCOME COVERED

CO1: Apply the principle of the foundational concepts of ethical hacking, including types of hackers, ethical principles, legal frameworks, and the role of responsible disclosure in cybersecurity.

ASSIGNMENTS QUESTIONS

1. A rival security researcher discloses a vulnerability you found first, taking public credit, after you had already privately reported it to the vendor under an embargo. Discuss the ethical norms around disclosure credit, embargoes, and how researchers should handle disputes like this professionally.

Code of Conduct:

Dony Bharath D / 1924L260001 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Rubrics

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all data-handling requirements and operations	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Application of Concepts	30	Accurately applies appropriate data structures and algorithms to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct code, output and complexity analysis with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

ETHICAL DISCLOSURE OF VULNERABILITIES: CREDIT, EMBARGOES, AND PROFESSIONAL DISPUTES

1. Introduction

In ethical hacking, responsible disclosure means reporting security vulnerabilities to the affected company before sharing them publicly. Sometimes, another security researcher may disclose the same vulnerability first and take public credit, even though we reported it privately earlier. This situation raises ethical questions about fairness, professional honesty, and responsible security practices.

2. Ethical Norms Around Disclosure Credit

- Give proper credit: Researchers should acknowledge the person who discovered and reported the vulnerability first, when the facts support that claim.
- Be honest: No researcher should falsely claim another person's discovery as their own.
- Maintain evidence: Researchers should keep dated emails, vulnerability reports, and vendor acknowledgements to establish their contribution.
- Respect teamwork: If multiple researchers contributed, their individual efforts should be recognized fairly.
- Avoid false accusations: Researchers should verify the facts before publicly blaming another researcher.

3. Ethical Norms Around Embargoes

An embargo is an agreed period during which a vulnerability is kept confidential before public disclosure.

- Respect confidentiality: Researchers should follow the agreed disclosure timeline.

- Protect users: Keeping details private temporarily helps the vendor prepare a security patch.
- Follow the agreement: Researchers should not disclose technical details before the agreed date without a valid reason.
- Communicate with the vendor: If the vulnerability is disclosed early by someone else, the researcher should contact the vendor to clarify the situation.
- Prioritize public safety: Disclosure decisions should consider the risk to users and whether a fix is available.

4. How to Handle the Dispute Professionally

1. Collect evidence: Keep the original report, timestamps, emails, and vendor acknowledgements.
2. Contact the vendor privately: Explain the situation and request clarification about the disclosure timeline and researcher credit.
3. Contact the rival professionally: Ask for clarification without using threats, insults, or personal attacks.
4. Request proper recognition: If the evidence supports the claim, request acknowledgement in the advisory, security report, or publication.
5. Use official channels: If the dispute remains unresolved, follow the vendor's vulnerability disclosure or bug bounty dispute process.
6. Avoid harmful actions: Never retaliate by leaking confidential information, attacking systems, or publishing sensitive user data.
7. Learn from the incident: Improve communication and agree on clear disclosure and credit policies for future research.

5. Example Scenario

Suppose Researcher A discovers a vulnerability and privately reports it to a software company under an agreed embargo. Before the embargo ends, Researcher B publicly announces the vulnerability and claims credit.

Researcher A should collect the original report and vendor acknowledgement, contact the vendor privately, and request fair recognition. The vendor should investigate the timeline and clarify the facts. Both researchers should communicate professionally and avoid sharing sensitive details that could put users at risk.

6. Conclusion

Ethical vulnerability disclosure requires honesty, confidentiality, fairness, and professional communication. Researchers should receive appropriate credit for their contributions, and embargo agreements should be respected. When disputes arise, they should be resolved through evidence, discussion, and official procedures rather than public arguments or retaliation. Protecting users and maintaining trust in the cybersecurity community must remain the highest priorities.